

Yao Ji

H. Milton Stewart Postdoctoral Fellow

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Postdoc

Georgia Institute of Technology, Atlanta, Georgia

Oct. 2024–Present

H. Milton Stewart Postdoctoral Fellow

H. Milton Stewart School of Industrial and Systems Engineering (ISyE)

Mentor: Guanghui (George) Lan

Current focus: Parameter-free stochastic optimization; variational inequalities; gradient norm minimization; high-order optimization; reinforcement learning; reinforcement learning in healthcare.

Education

Purdue University, West Lafayette, Indiana

Aug. 2019–Aug. 2024

Ph.D. in Operations Research

Committee: Harsha Honnappa (advisor), Raghu Pasupathy, Alex Wang, and Gesualdo Scutari (advisor)

Dissertation: *High-Dimensional Inference over Networks: Statistical and Computational Guarantees*

Beijing Normal University, Beijing, China

Aug. 2016–Jun. 2019

M.S. in Probability and Mathematical Statistics

advisor: Wenming Hong

Thesis: *Conditional Limit Theorem for Bellman–Harris Branching Process*

Beijing Normal University, Beijing, China

Aug. 2012–Jun. 2016

B.S. in Mathematics

Thesis: *Conceptual New Proofs of Geometric Convergence of Moment Generating Function for Galton–Watson Process in the Noncritical Case*

Research Interests

Stochastic optimization; reinforcement learning; healthcare.

Research Papers

JOURNAL PAPERS

Yao Ji, Guanghui Lan, and Jason Zhu

Computation of Strong Solutions to Stochastic Variational Inequalities

Journal manuscript | Under review, *Mathematical Programming* — first-round review; [arXiv:2609.04188](#) (2026).

Yao Ji and Guanghui Lan

Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates

Job-market paper | Journal manuscript | Under review, *Mathematical Programming* — first-round review; [arXiv:2604.06525](#) (2026).

Yao Ji and Guanghui Lan

High-order Accumulative Regularization for Gradient Minimization in Convex Programming

Journal manuscript | R&R, *SIAM Journal on Optimization* — second-round review; [arXiv:2511.03723](#) (2025).

Yao Ji, Gesualdo Scutari, Ying Sun, and Harsha Honnappa

Distributed Sparse Regression via Penalization

Journal article | Published, *Journal of Machine Learning Research*, 24(272):1–62, 2023. [Paper](#).

Yao Ji, Gesualdo Scutari, Ying Sun, and Harsha Honnappa

Distributed (ATC) Gradient Descent for High Dimension Sparse Regression

Journal article | Published, *IEEE Transactions on Information Theory*, 69(8):5253–5276, 2023. [Paper](#).

Vladimir A. Vatutin, Wenming Hong, and **Yao Ji**
Reduced critical Bellman–Harris branching processes for small populations
 Journal article | Published, *Discrete Mathematics and Applications*, 28(5):319–330, 2018. [Paper](#).

CONFERENCE AND WORKSHOP PAPERS

Ruqi Bai, **Yao Ji**, Mingyu Kim, Easton Currie, Zeyu Zhou, and David I. Inouye
Provable Robustness to Spurious Correlations via Invariant Data for Robust Finetuning
 Workshop paper | Accepted, AISTATS 2026 Workshop on Causality in the Age of AI Scaling. [Paper](#).

Ruqi Bai, **Yao Ji**, Zeyu Zhou, and David I. Inouye
From Invariant Representations to Invariant Data: Provable Robustness to Spurious Correlations via Noisy Counterfactual Matching
 Preprint | [arXiv:2505.24843](#) (2025).

WORK IN PROGRESS

Yao Ji, Yan Li, Guanghui Lan, Zehua Dong, Shuoling Li, Ilker Guven, Xiaoyu Chen, Lihui Bai, and Jiapeng Huang
Transformer-Based Reinforcement Learning for Perioperative Management of Cardiac Surgery-Associated Acute Kidney Injury

Yao Ji, Ruqi Bai, and Guanghui Lan
Benchmarking Stochastic Auto-conditioned Fast Gradient Methods

Teaching Experience

Georgia Institute of Technology, ISyE	
Advisor	
ISyE 4106: Senior Design	Fall 2026
Purdue University, Industrial Engineering	
Teaching Assistant	
IE 335: Operations Research	Spring and Fall 2023
IE 330: Probability and Statistics in Engineering	Fall 2022
IE 590: Introduction to Optimization Algorithms (graduate-level)	Fall 2022
Beijing Normal University, School of Mathematical Sciences	
Co-Lecturer	
Large Deviation Theory	Spring 2018
Brownian Motion	Fall 2017
Random Walk in Random Environment	Spring 2017
Galton–Watson Branching Process	Fall 2016
Teaching Assistant ; Outstanding Teaching Assistant Award recipient	
Measure Theory I	Fall 2017 and Fall 2018
Measure Theory II	Spring 2017 and Spring 2018
Stochastic Calculus for Finance (graduate-level)	Fall 2016

Outreach

<i>Instructor</i> : Pre-College Reinforcement Learning Course for High School Students, Georgia Tech	Summer 2026
<i>Judge</i> : STEM Spark: INFORMS K–12 Poster Competition	2026
<i>Judge</i> : Georgia Tech 2026 CRIDC Poster Competition	2026

Awards and Honors

H. Milton Stewart Postdoctoral Fellowship, Georgia Institute of Technology	2024
Lee A. Chaden Fellow IE Scholarship, Purdue University	2023
Graduate School Summer Research Grant, Purdue University	2023
Travel Grant, School of Industrial Engineering, Purdue University	2022, 2023
Ross Fellowship, Purdue University	2019, 2020
Dr. Theodore J. and Isabel M. Williams Fellowship in Industrial Control Systems, Purdue University	2019
First Prize Scholarship (2nd of 53, School of Mathematics), Beijing Normal University	2018
First Prize Scholarship (1st of 12, Markov Process), Beijing Normal University	2017
Outstanding Teaching Assistant for Measure Theory, Beijing Normal University	2017
Outstanding Undergraduate Thesis, School of Mathematics, Beijing Normal University	2016
First Prize Scholarship (top 5%), School of Mathematics, Beijing Normal University	2015
Second Prize Scholarship, Beijing Normal University	2014
Second Prize, China Undergraduate Mathematical Modeling Contest (top 5%)	2014

Talks

<i>Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates</i>	
– MOPTA, Lehigh University, Bethlehem, PA	Aug. 2026
– Clemson University, School of Mathematical and Statistical Sciences, Clemson, SC	Apr. 2026
– INFORMS Optimization Society Conference, Atlanta, GA	Mar. 2026
<i>High-order Accumulative Regularization for Gradient Minimization in Convex Programming</i>	
– INFORMS Annual Meeting, Atlanta, GA	Oct. 2025
<i>Distributed (ATC) Gradient Descent for High Dimension Sparse Regression</i>	
– International Symposium on Mathematical Programming (ISMP), Montréal, Canada	Jul. 2024

Poster Presentations

<i>Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates</i>	
– Southern California OR/OM Day, USC Marshall, Los Angeles, CA	May 2026
<i>Transformer-Based Patient Response Modeling Across All Perioperative Phases for Cardiac Surgery-Associated Acute Kidney Injury Management</i>	
– Society of Cardiovascular Anesthesiologists Annual Meeting, Nashville, TN	Apr. 2026
<i>Distributed (ATC) Gradient Descent for High Dimension Sparse Regression</i>	
– SIAM Conference on Mathematics of Data Science, Atlanta, GA	Oct. 2024
– Midwest Machine Learning Symposium, University of Illinois Chicago, Chicago, IL	May 2023
– Statistics and Optimization in Data Science Workshop, Purdue University, West Lafayette, IN	May 2023
<i>Distributed Sparse Regression via Penalization</i>	
– Cornell ORIE Young Researchers Workshop, Ithaca, NY	Oct. 2023
– International Conference on Continuous Optimization (ICCOPT)/MOPTA, Lehigh University, Bethlehem, PA	Jul. 2022

Service

Journal Reviewer: *Mathematical Programming*; *SIAM Journal on Optimization*; *Operations Research*; *Journal of the American Statistical Association*; *Stochastic Systems*; *Journal of Global Optimization*; *Journal of Optimization Theory and Applications*; and *IEEE Transactions on Automatic Control*.

Conference Reviewer: IEEE International Symposium on Information Theory.